# Embedded Agent Controller (EAC)

### *Project Architecture & System Overview*

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## 1. Project Overview

EAC (Embedded Agent Controller) is a deterministic, schema-governed intent mediation engine that translates natural language input into validated, structured action instructions. EAC does NOT execute business logic. It does NOT mutate application state. It acts as a governance layer between AI reasoning and system execution.

## 2. Repository Structure

```
EAC/
■
■■■ frontend/     → Next.js user-facing application
■
■■■ backend/
    ■■■ internal/
        ■■■ core/
            ■■■ api/        → HTTP transport layer
            ■■■ engine/     → Intent orchestration logic
            ■■■ llm/        → AI provider abstraction
            ■■■ registry/   → Action schema management
            ■■■ validation/ → Deterministic schema validation
            ■■■ policy/     → Feature & entitlement enforcement
            ■■■ audit/      → Audit logging
            ■■■ storage/    → Database access layer
            ■■■ models/     → Domain models
            ■■■ contracts/  → Public request/response structs
```

## 3. High-Level Architecture

```
User → Frontend (Next.js)
     → Backend API
     → Agent Engine
         → Load Action Schemas
         → Build Structured Prompt
         → Call LLM Provider
         → Validate Structured Output
         → Enforce Policy
```

```
    → Write Audit Record
→ Return Structured Instruction
→ Host Application Executes (Optional)
```

## 4. Core Responsibilities

- Schema-first action modeling
- LLM abstraction (Groq, OpenAI, etc.)
- Deterministic JSON validation
- Feature gating & entitlement checks
- Audit trail generation
- Structured action output guarantee

## 5. Non-Responsibilities

- Business logic execution
- UI navigation control
- Database mutation in host system
- Financial transaction handling
- Domain state validation

## 6. Core Data Models

```
ActionModel:
- ID
- OrgID
- Name
- Description
- Parameters (JSON)
- Rules (JSON)
- RequiredFeature
- Version
- CreatedAt

AuditRecord:
- ID
- OrgID
- UserID
- Prompt
- ProposedAction (JSON)
- Validated (bool)
- FinalResponse (JSON)
```

```
- CreatedAt
```

## 7. Security Model

- Strict JSON schema validation
- Unknown field rejection
- Enum enforcement
- No arbitrary execution
- No dynamic evaluation
- Structured LLM output only

## 8. Summary

EAC provides a deterministic boundary between AI reasoning and system authority. It ensures governance, auditability, and schema-constrained execution while allowing the host application to retain full execution control.